

Carbohydrates

Organic Chemistry II

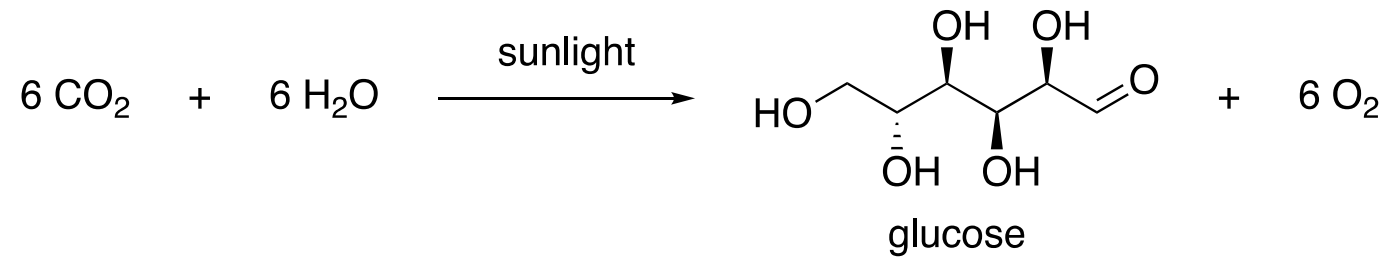
CHEM 2312

Dr. Michael Evans

- Recognize the general structures and typical biochemical roles of carbohydrates.
- Classify monosaccharides as aldoses or ketoses and assign the stereochemical descriptors D and L.
- Determine isomeric relationships between structurally related aldoses and ketoses.
- Draw structures for cyclic monosaccharides, determining and distinguishing between α and β anomers.
- Predict the products of, and draw plausible mechanisms for, mutarotation of monosaccharides.
- Predict the products of, and draw plausible mechanisms for, typical reactions of monosaccharides, including:
 - Etherification and acylation
 - Glycoside formation
 - Epimerization
 - Reduction and oxidation
- Design syntheses focused on chain lengthening (Fischer-Kiliani) or shortening (Wohl) of monosaccharides.
- Draw and interpret structures of disaccharides and polysaccharides.
- Recognize sugar units in derivatives such as amino sugars and *N*-glycosides (nucleic acids, etc.).

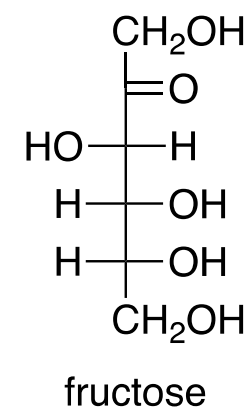
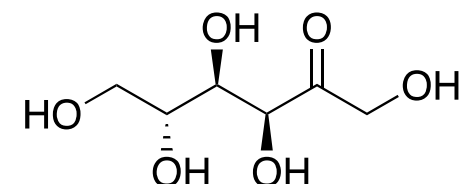
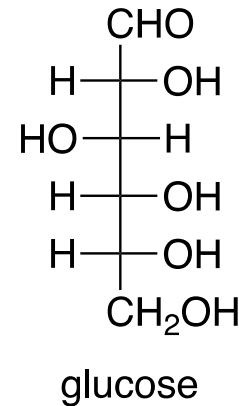
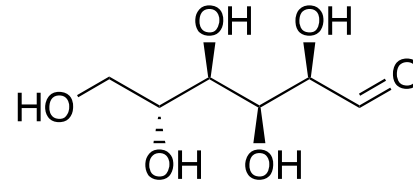
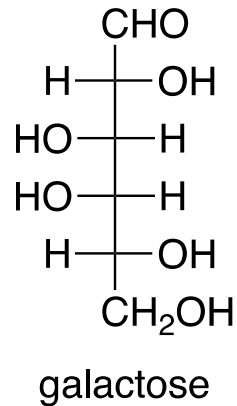
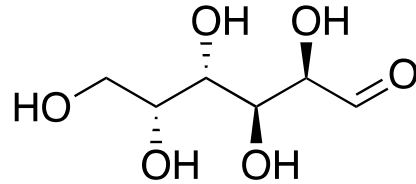
24.1 Introduction to Carbohydrates

- **Carbohydrates** are polyhydroxy ketones or aldehydes with a variety of biochemical functions, most notably energy storage and transduction and structural rigidity
- Photosynthesis converts solar energy into chemical energy in the form of glucose
- Animals release this energy by converting glucose back into CO_2 and H_2O



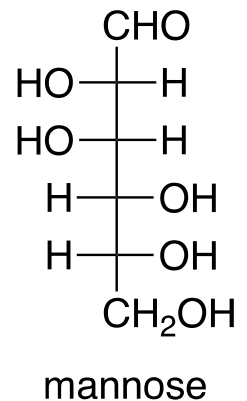
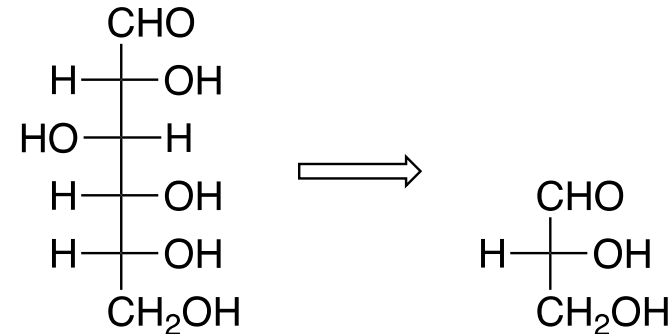
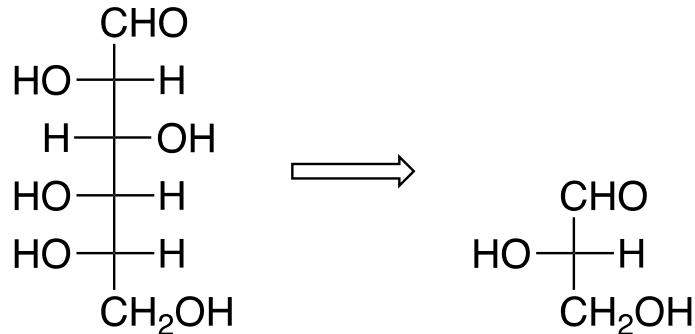
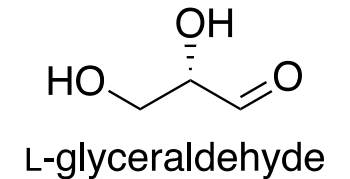
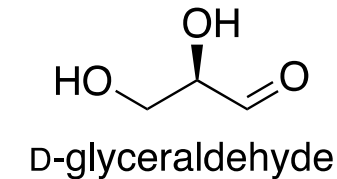
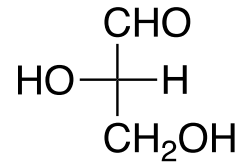
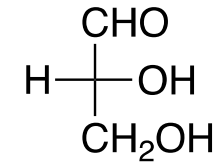
24.2 Classifying Carbohydrates by Functional Group

- **Monosaccharides** contain a single sugar unit (typically one carbonyl group or equivalent)
- **Aldoses** are monosaccharides containing an aldehyde group; **ketoses** contain a ketone group
- A Greek root indicating the number of carbons in the sugar is commonly added between “aldo/ket” and “ose”



24.2 Classifying Enantiomers of Monosaccharides

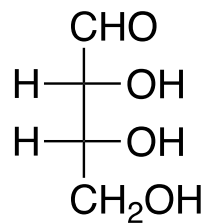
- Enantiomeric monosaccharides differ in configuration at all stereocenters, beginning with the triose **glyceraldehyde**
- Larger aldoses can be degraded to glyceraldehyde via removal of one carbon at a time, starting with the aldehyde carbon
 - All that degrade to D-glyceraldehyde are called **D-sugars**
 - All that degrade to L-glyceraldehyde are called **L-sugars**



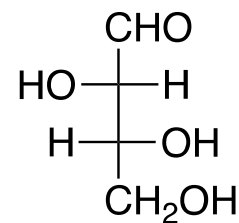
24.3 A Survey of Aldoses | 1 of 2



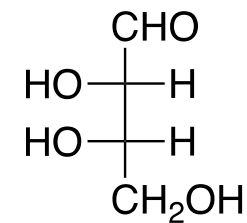
- There are two diastereomeric **aldotetroses**, each of which has an enantiomer



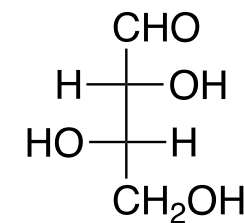
D-erythrose



D-threose

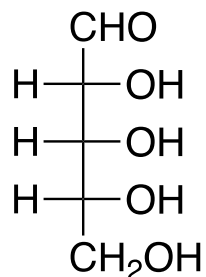


L-erythrose

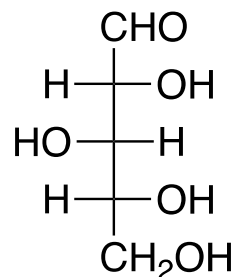


L-threose

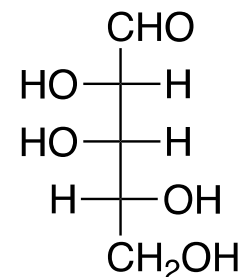
- There are four diastereomeric **aldopentoses**, each of which has an enantiomer



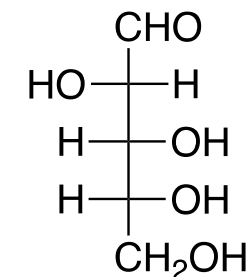
D-ribose



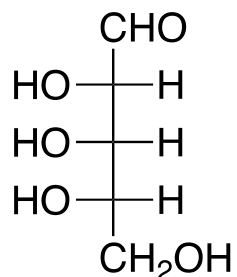
D-xylose



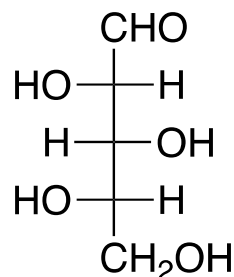
D-lyxose



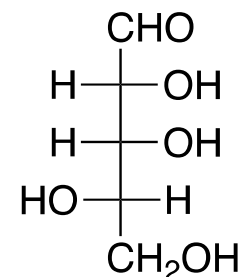
D-arabinose



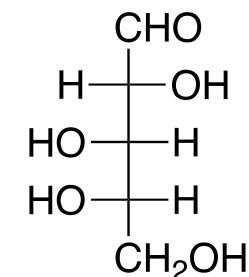
L-ribose



L-xylose

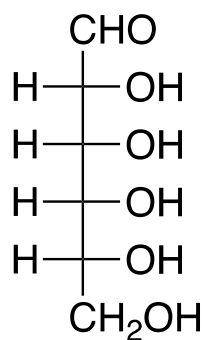


L-lyxose

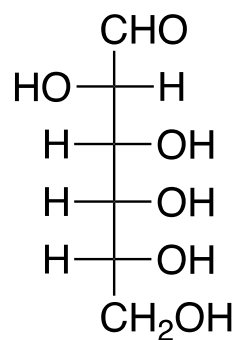


L-arabinose

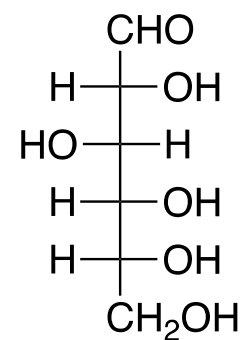
- There are eight diastereomeric **aldohexoses**, each of which has an enantiomer



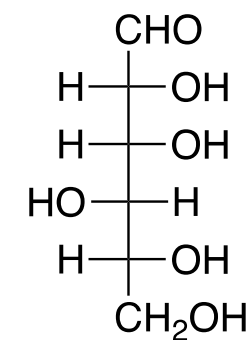
D-allose



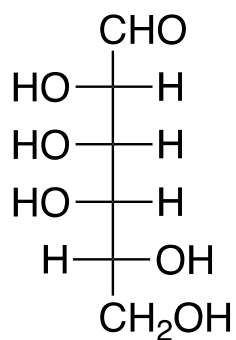
D-altrose



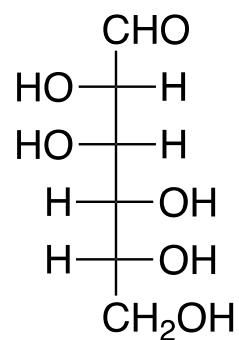
D-glucose



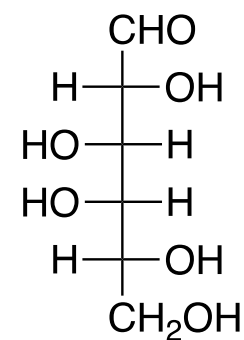
D-gulose



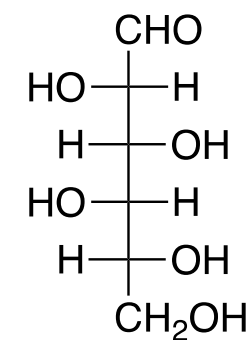
D-talose



D-mannose



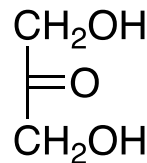
D-galactose



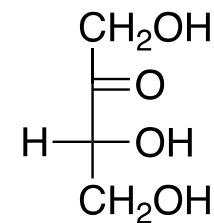
D-idose

24.4 A Survey of Ketoses

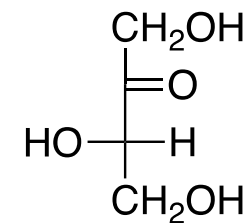
- The only **ketotriose** is achiral dihydroxyacetone; there are two enantiomeric **ketotetroses**



dihydroxyacetone

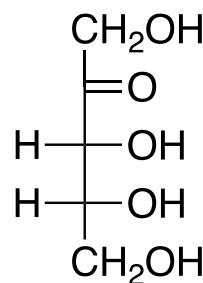


D-erythrulose

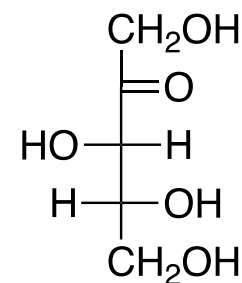


L-erythrulose

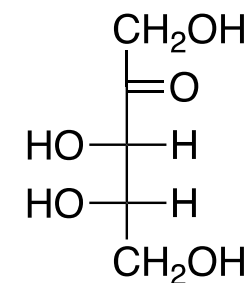
- There are two diastereomeric **ketopentoses**, each of which has an enantiomer



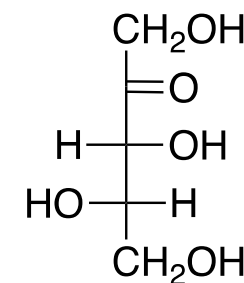
D-ribulose



D-xylulose

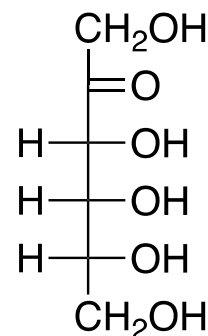


L-ribulose

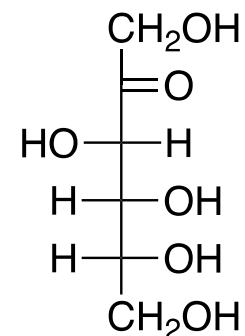


L-xylulose

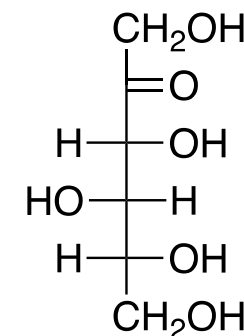
- There are four diastereomeric **ketoheptoses**, each of which has an enantiomer



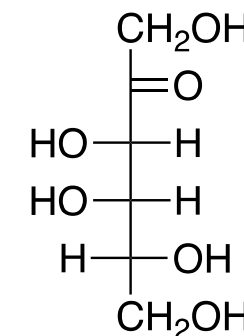
D-psicose



D-fructose



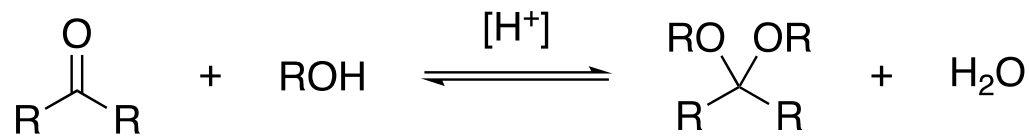
D-sorbose



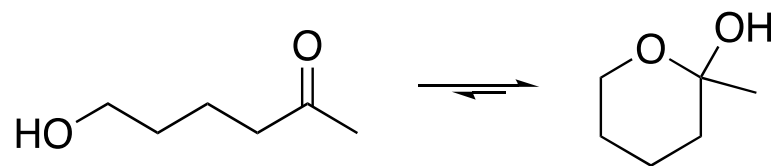
D-tagatose

24.5 Cyclization of Hydroxyaldehydes: Hemiacetals Revisited

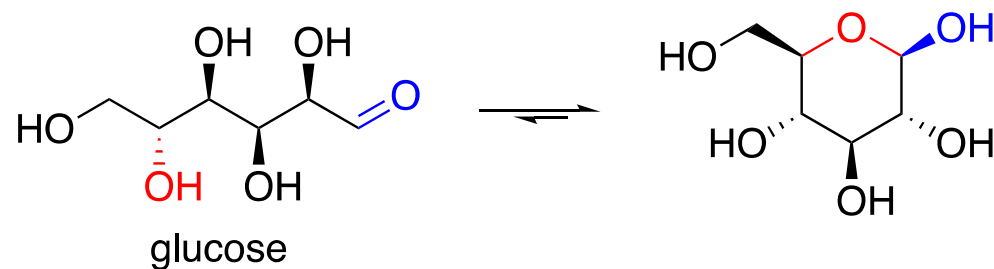
- Ketones and aldehydes react with alcohols in the presence of acid to form acetals



- Hydroxyaldehydes and hydroxyketones can cyclize readily to form cyclic hemiacetals

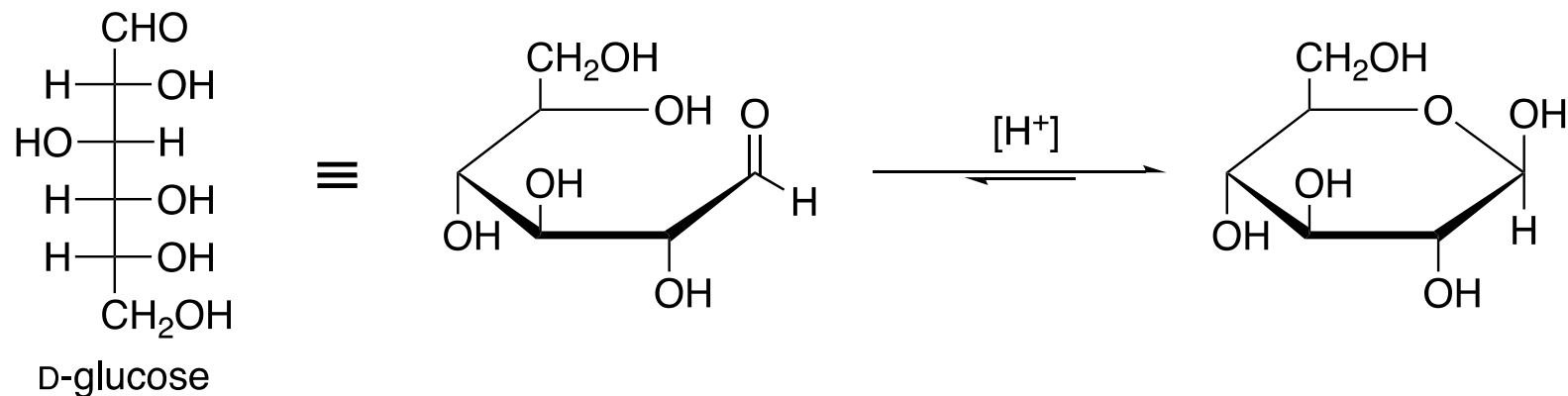


- Carbohydrates can undergo this reaction and tend to exist mostly in their cyclic forms



24.5 Pyranose Forms of Glucose

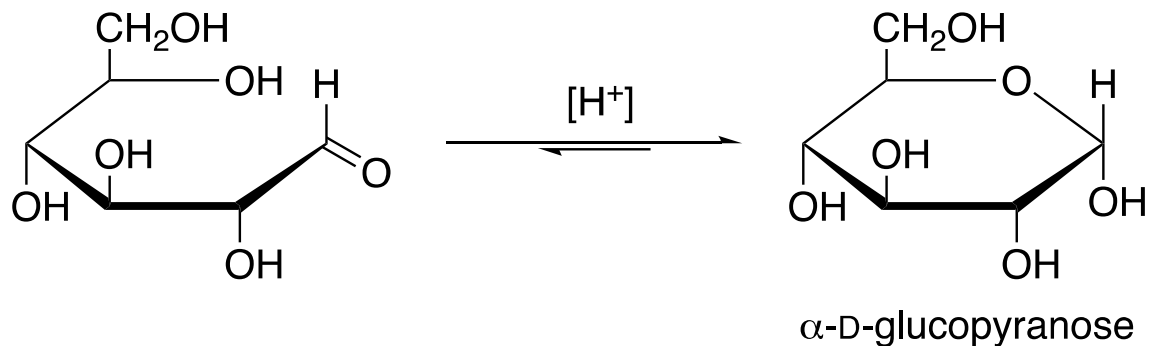
- Glucose cyclizes via the C-5 hydroxyl group to form a six-membered **pyranose** ring
- At equilibrium, the cyclic or closed pyranose form is *extremely* favored (>99.99%)



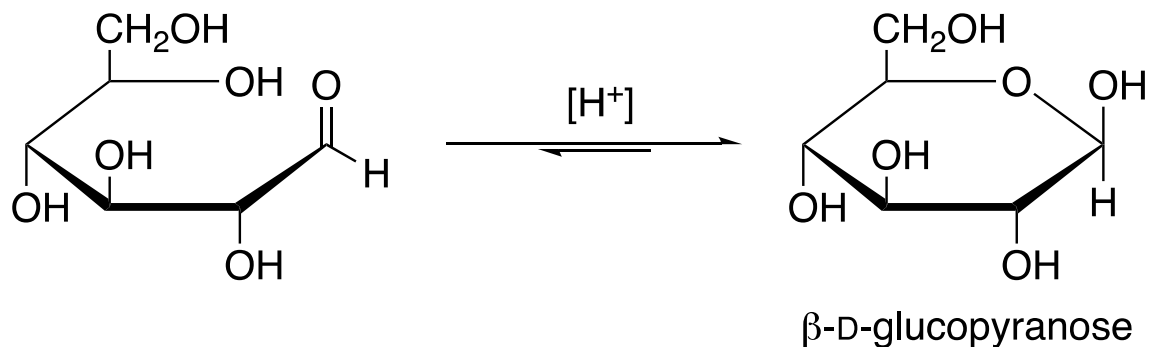
- Cyclization is typically catalyzed by acid or base and follows the mechanism of hemiacetal formation we have previously seen (1. pt; 2. Ad_N; 3. pt)

24.5 α - and β -Anomers Differ in Configuration at the Hemiacetal Carbon (C-1)

- Cyclization establishes a new stereocenter at C-1, leading to diastereomers known as **anomers**
- In the α anomer, the new -OH group points in the same direction as the -OH at the “bottom” stereogenic carbon

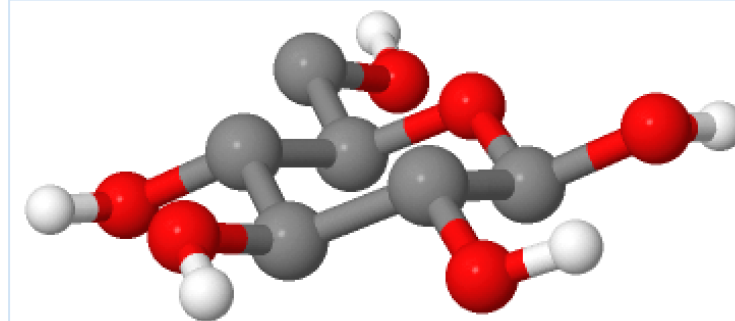
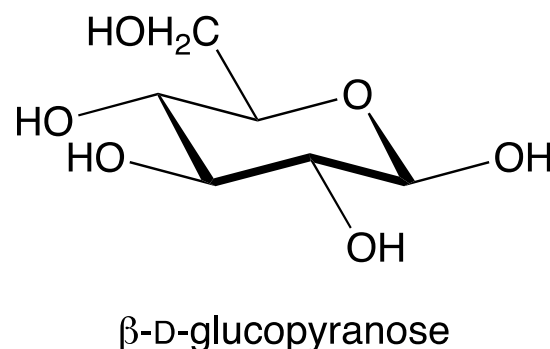
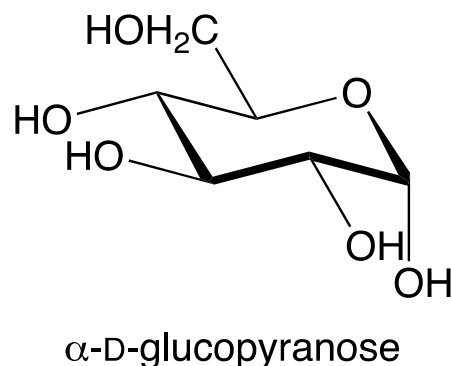
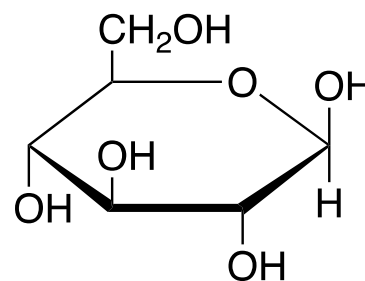
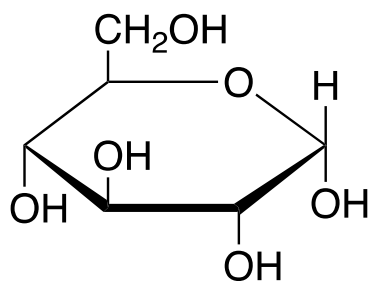


- In the β anomer, the new -OH group points in the opposite direction of the -OH at the “bottom” stereogenic carbon



24.5 Representing Pyranoses in Three Dimensions

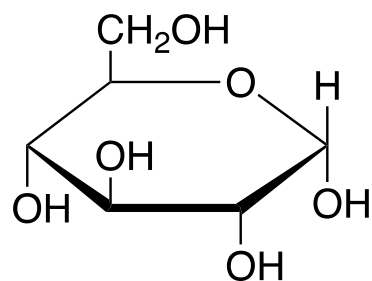
- Pyranoses may be represented as Fischer projections or line-angle projections in two dimensions
- In three dimensions, **Haworth projections** or **chair conformers** may be drawn



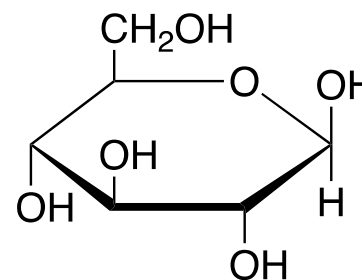
The beta anomer of D-glucose is entirely free of strain and has only equatorial hydroxyl groups.

24.5 Physical Properties of Anomers and Mutarotation

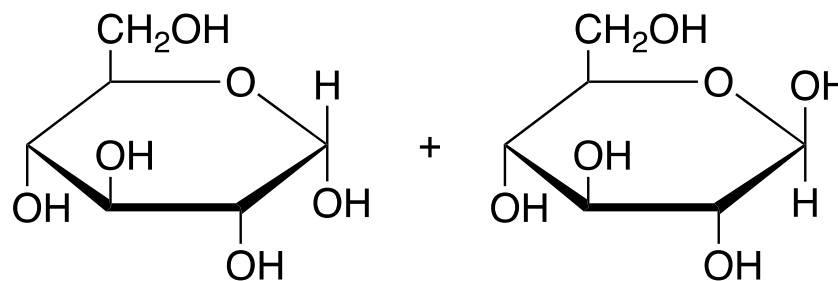
- Anomers are diastereomers and thus have different physical and chemical properties
- For example, specific rotations of the pure anomers differ
- However, they converge to a common value of $+52.6^\circ$ over time via a process called **mutarotation**



α -D-glucopyranose
 $[\alpha]_D = +112.2^\circ$

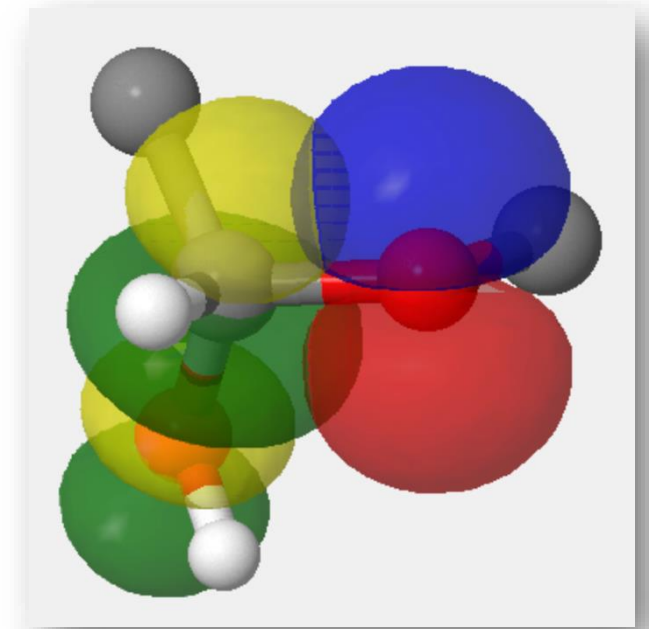
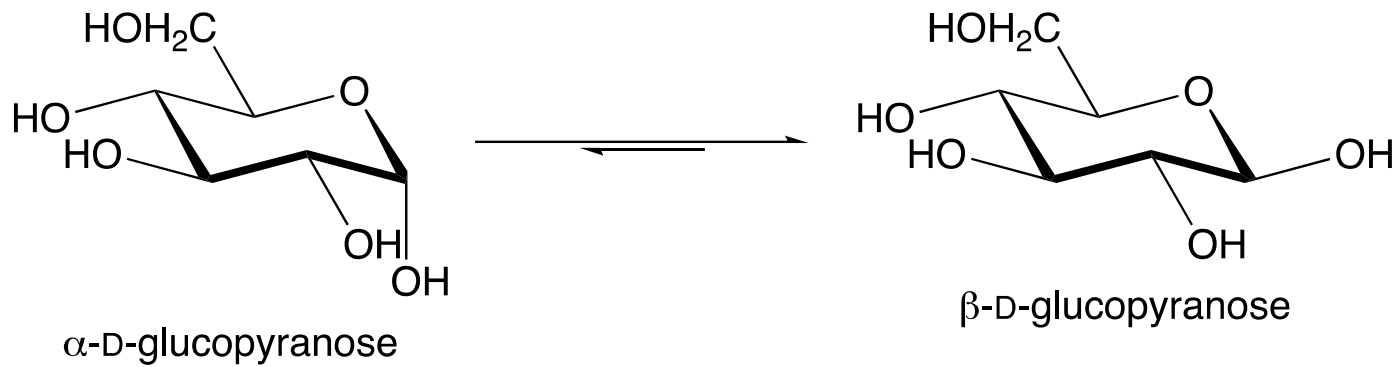


β -D-glucopyranose
 $[\alpha]_D = +18.7^\circ$



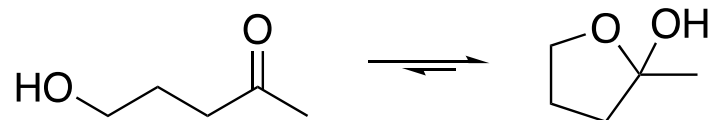
equilibrium mixture of anomers (33:67)
 $[\alpha]_D = +52.6^\circ$

- The α anomer contains an axial $-\text{OH}$ group but is more stable than expected from simple free-energy calculations
- This stabilization is explained by an $n \rightarrow \sigma^*$ orbital interaction (the **anomeric effect**)

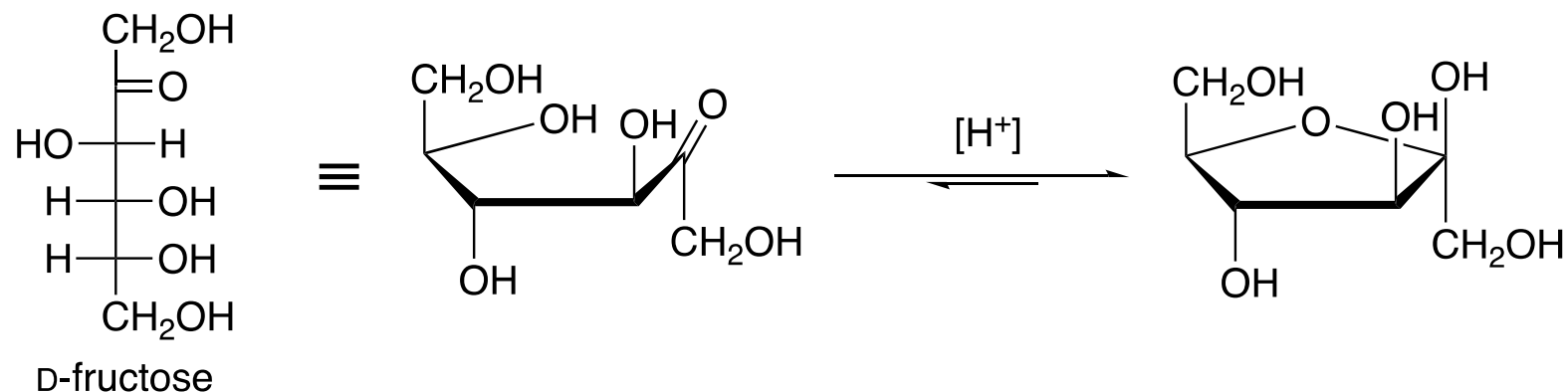


24.5 Furanose Forms of Monosaccharides

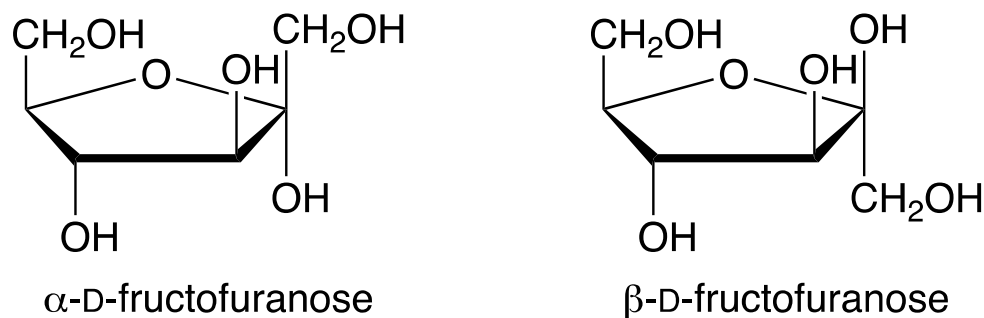
- Hydroxy aldehydes and hydroxy ketones can also cyclize to form five-membered rings



- Five-membered cyclic monosaccharides are known as **furanoses**

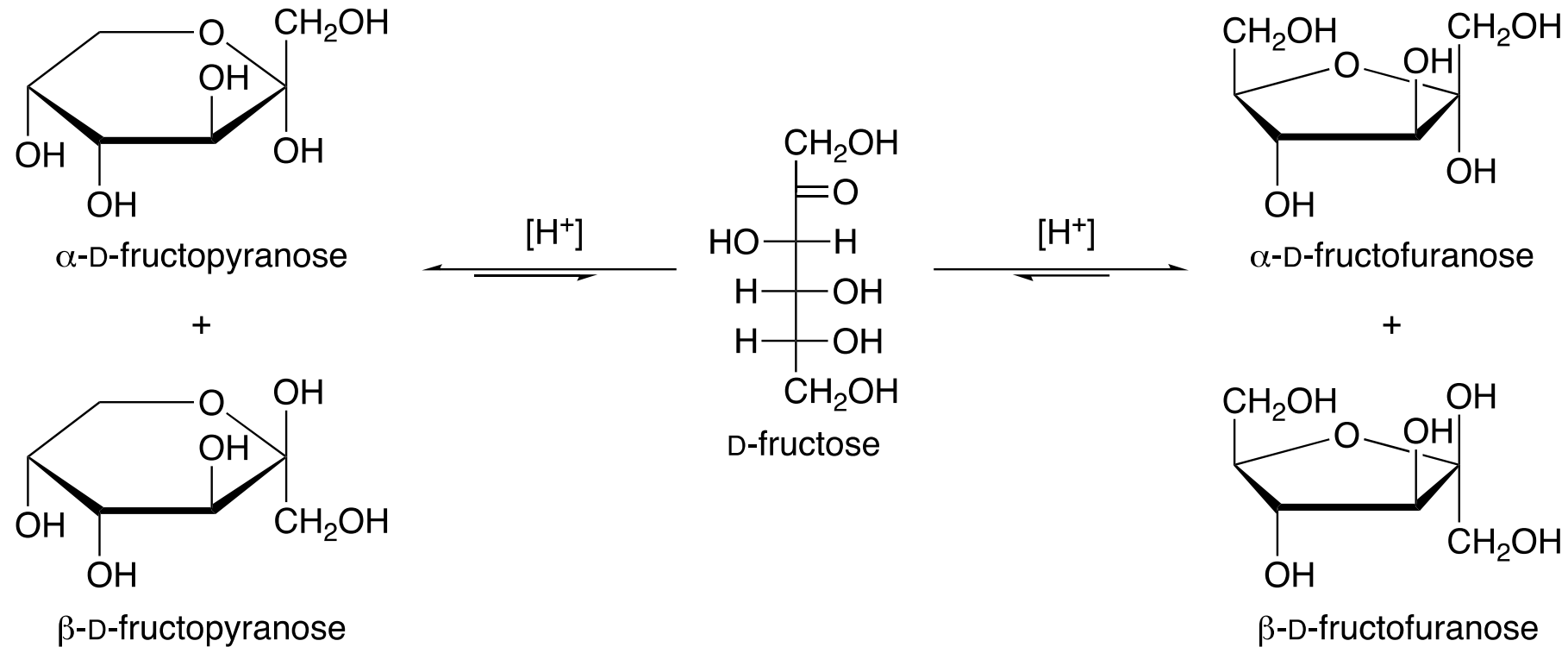


- Two anomers can be identified for furanoses as well



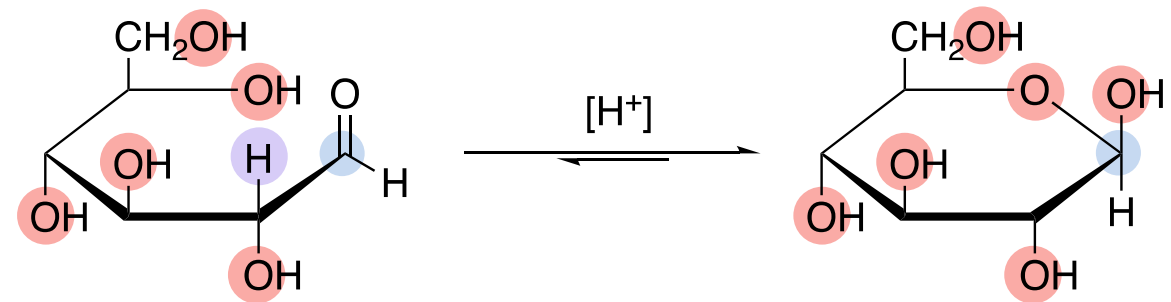
24.5 Fructose is a Mixture of Furanose and Pyranose Forms

- At equilibrium, fructose is a mixture of pyranoses and furanoses with very little open form



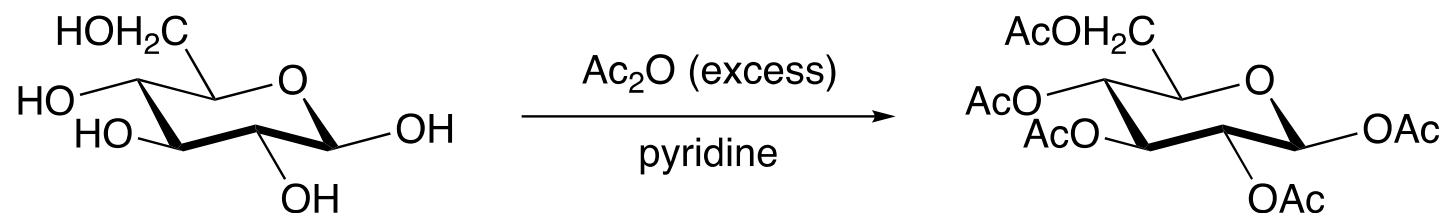
24.6 Reactions of Monosaccharides

- Like all biochemical monomers, monosaccharides contain nucleophilic and electrophilic groups in the same molecule
 - The hydroxyl (-OH) groups are **nucleophilic**
 - The anomeric carbon (C-1) is **electrophilic**
- **Nucleophilic reactivity of the hydroxyl groups**
 - Acylation (ester formation) and ether formation
 - Polymerization
 - Cyclization (hemiacetal formation)
- **Electrophilic reactivity of the anomeric carbon**
 - Nucleophilic addition
 - Glycoside formation
 - Reduction
- **Acidic reactivity of carbonyl alpha carbons**
 - Epimerization
 - Aldol and related condensations

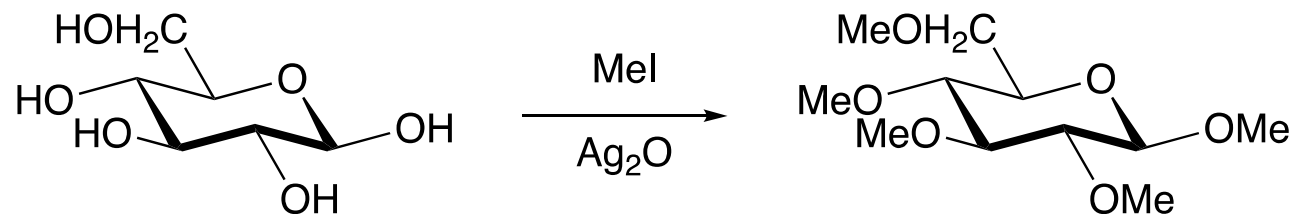


24.6 Esterification and Etherification of Hydroxyl Groups

- Substitution reactions involving the hydroxyl groups are common ($-\text{OH} \rightarrow -\text{OR}$)
- Exhaustive acylation converts all hydroxyl groups to esters, a nucleophilic acyl substitution ($\text{S}_{\text{N}}\text{Ac}$) reaction

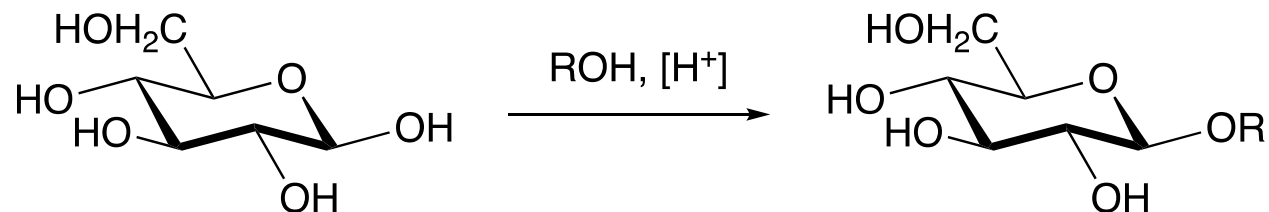


- Exhaustive etherification converts all hydroxyl groups to ethers, a Williamson ether synthesis ($\text{S}_{\text{N}}2$)

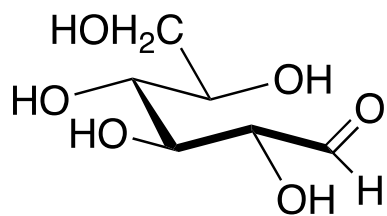
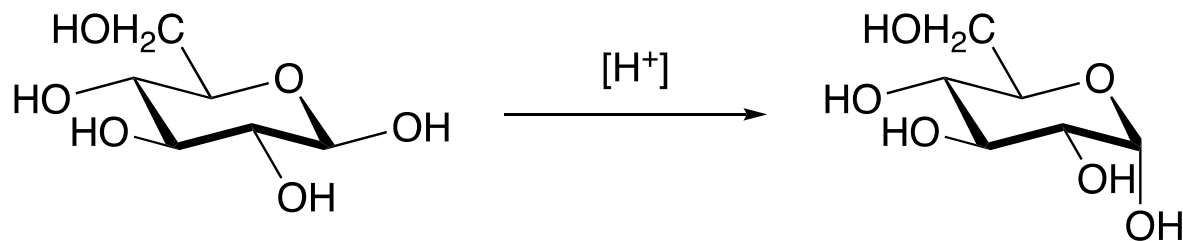


24.6 Glycoside Formation

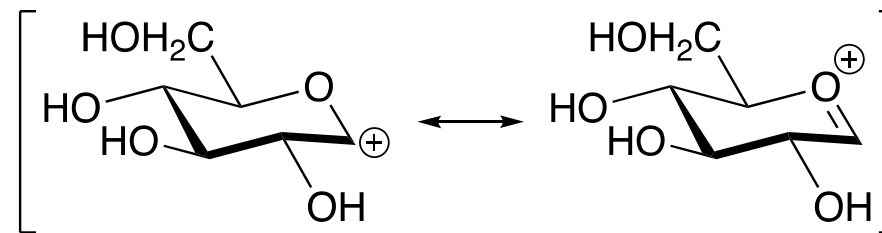
- Substitution at the anomeric carbon (C-**OH** → C-**Nu**) is mechanistically analogous to acetal formation
- When the nucleophile is an alcohol, the resulting “sugar acetal” is known as a **glycoside**
- Why aren't the other hydroxyl groups also replaced? Consider the mechanism...



- The mechanism of mutarotation can involve equilibration through the open form (path **A**) or loss of water to form an oxygen-stabilized carbocation followed by nucleophilic addition (path **B**)



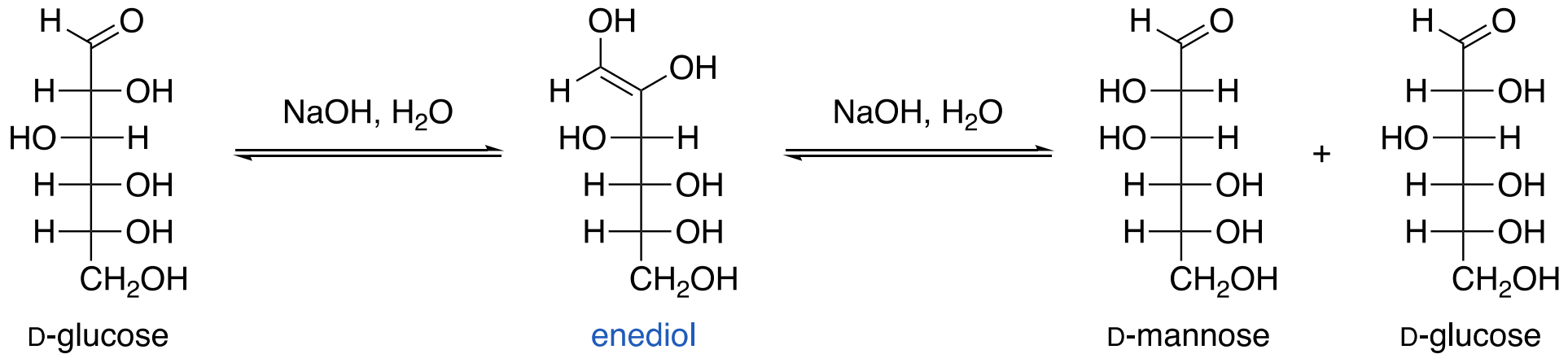
Path **A**



Path **B**

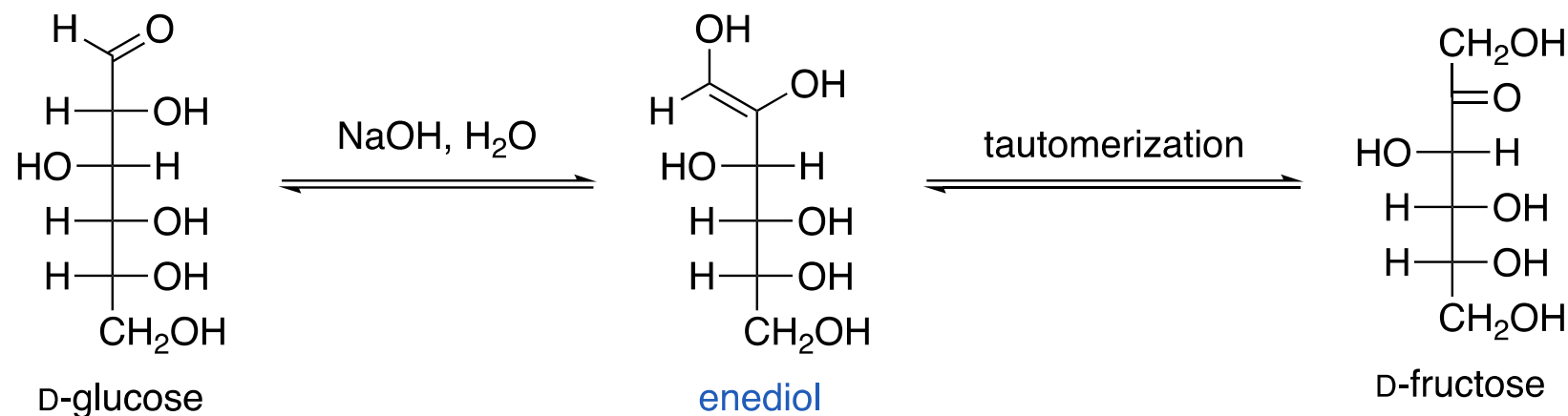
24.6 Epimerization

- In the presence of aqueous hydroxide base, glucose slowly equilibrates to a mixture of glucose and mannose
- The configuration at C-2 scrambles via an **enediol** intermediate
- Glucose and mannose, which differ in configuration only at one carbon (C-2), are called **epimers**



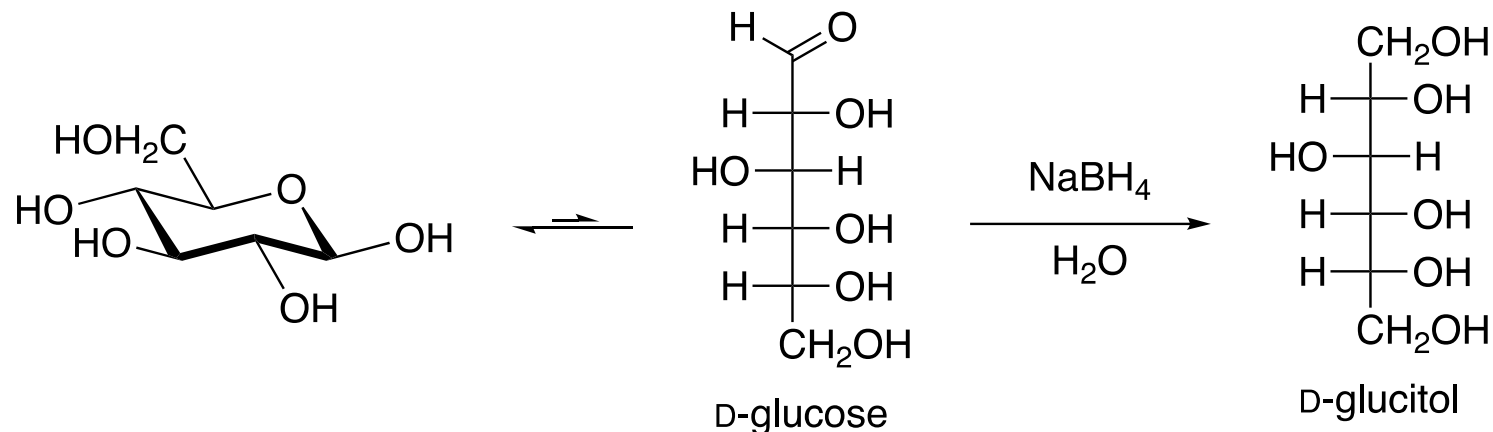
24.6 Migration of the Carbonyl Group (Isomerization)

- Re-protonation of the enediol intermediate at C-1 can result in migration of the carbonyl group to C-2, converting an aldose to a ketose (or *vice versa*!)

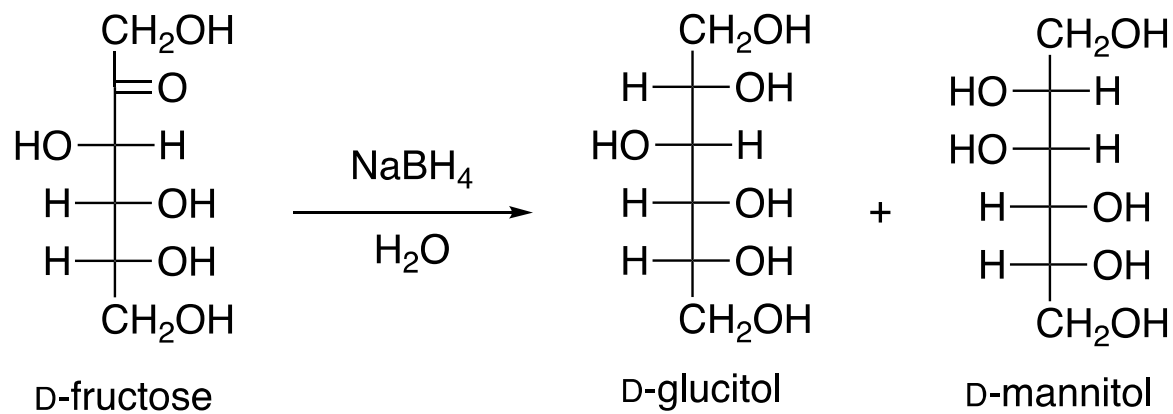


24.6 Reduction of Monosaccharides

- Reduction of ketoses or aldoses with sodium borohydride (NaBH_4) yields **alditols**

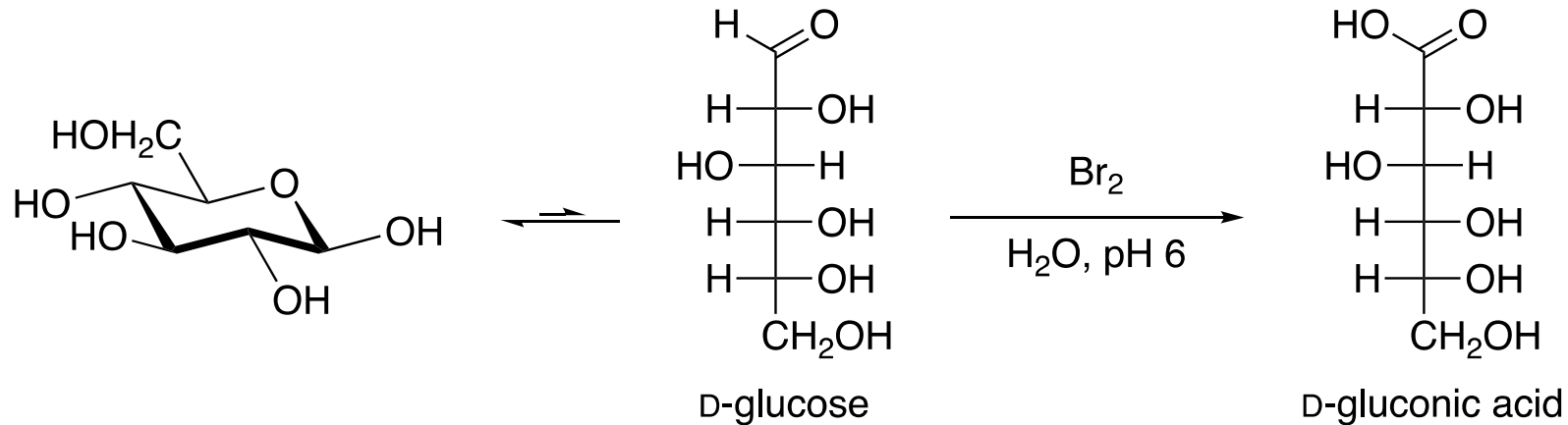


- Reduction of a ketose may yield a mixture of stereoisomeric alditols

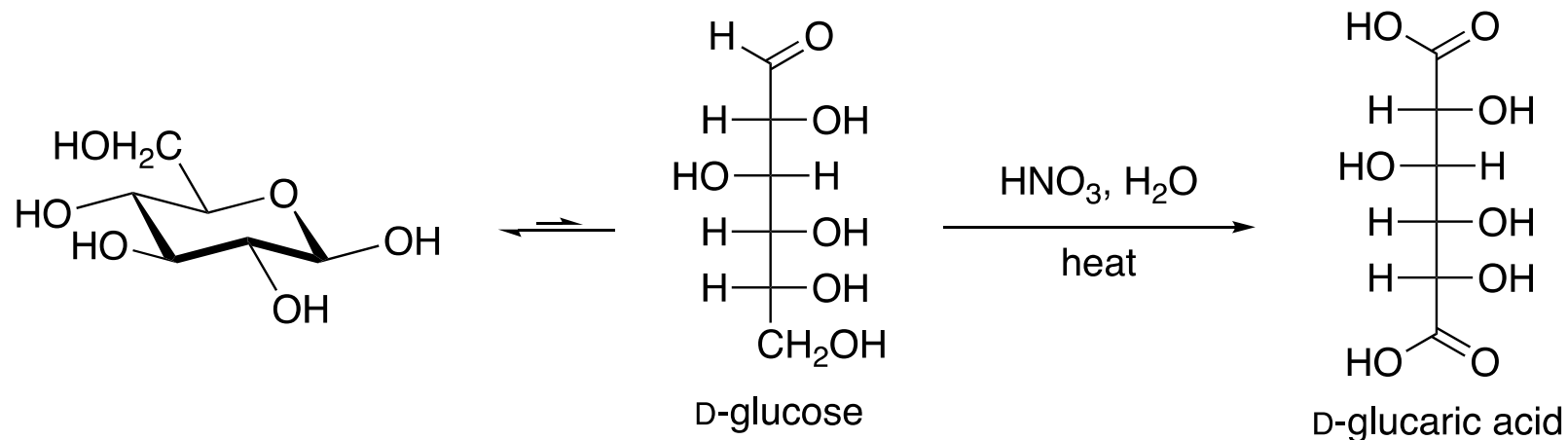


24.6 Oxidation of Monosaccharides

- “Bromine water” promotes oxidation of the aldehyde group of aldoses to a carboxylic acid, giving an **aldonic acid**

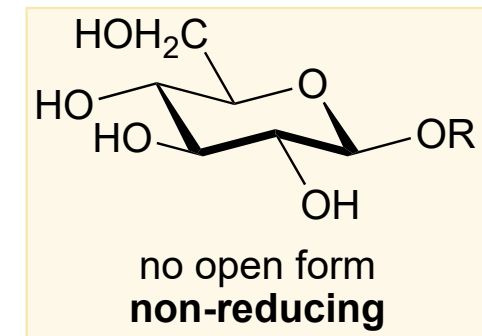
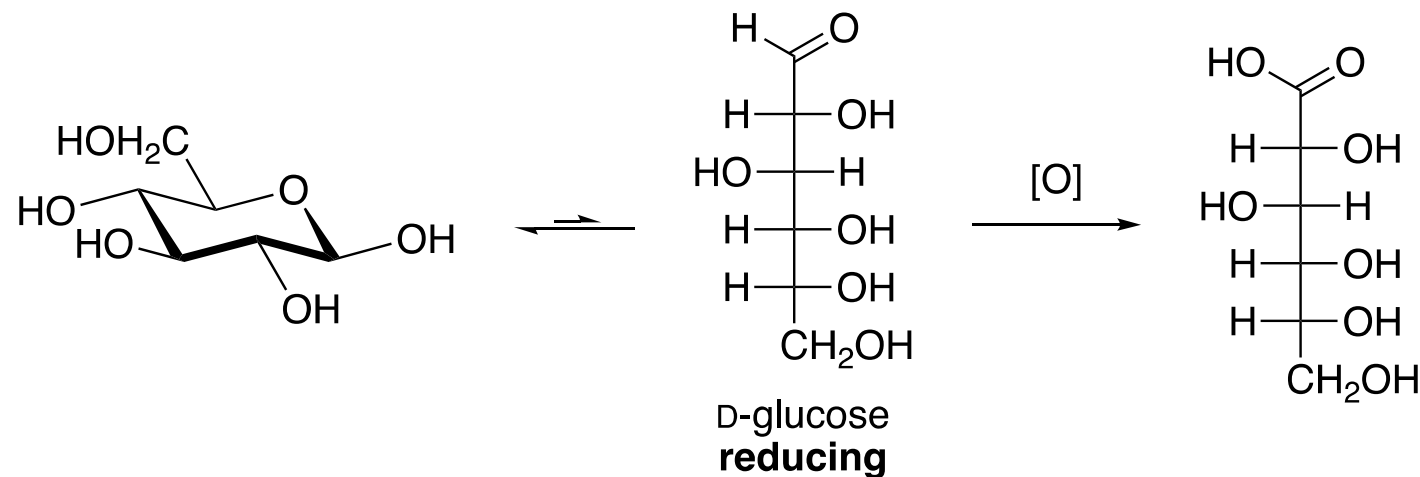


- Aqueous nitric acid is a stronger oxidizing agent that converts both the aldehyde and primary alcohol to carboxylic acid groups, giving an **aldaric acid**



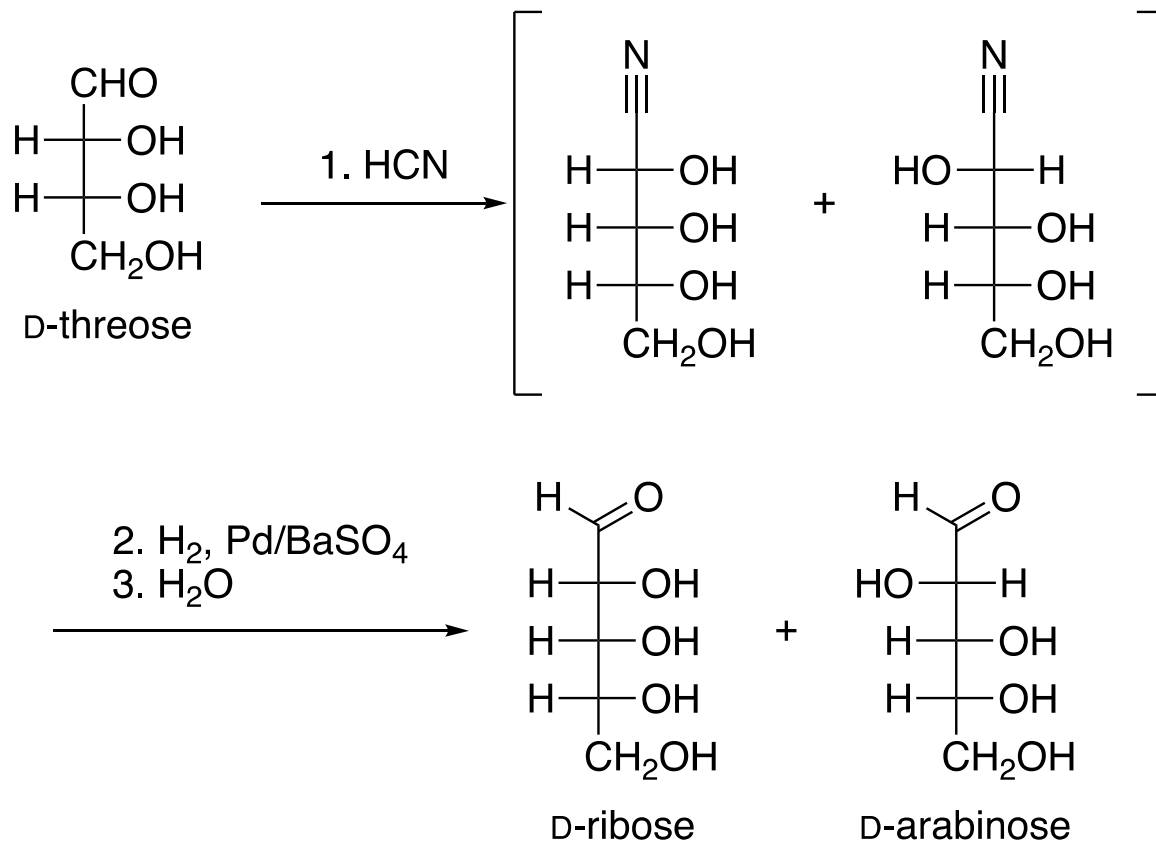
24.6 Reducing and Non-reducing Sugars

- Isomerization of ketoses to aldoses can occur under some oxidizing conditions
 - Tollens' reagent (Ag^+ in aqueous ammonia)
 - Fehling's reagent (Cu^{2+} in aqueous sodium tartrate)
 - Benedict's reagent (Cu^{2+} in aqueous sodium citrate)
- Sugars with a free aldehyde (aldoses or isomerized ketoses) in the open form are called **reducing sugars**
- Glycosides and related compounds substituted at C-1 are non-reducing



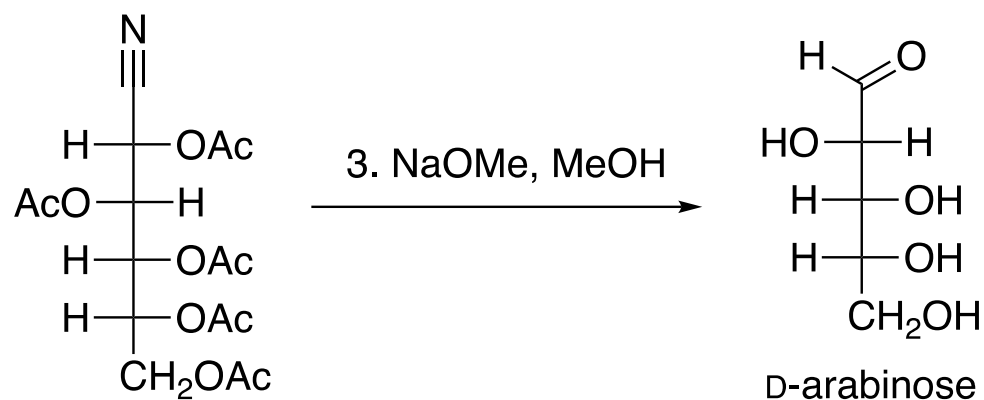
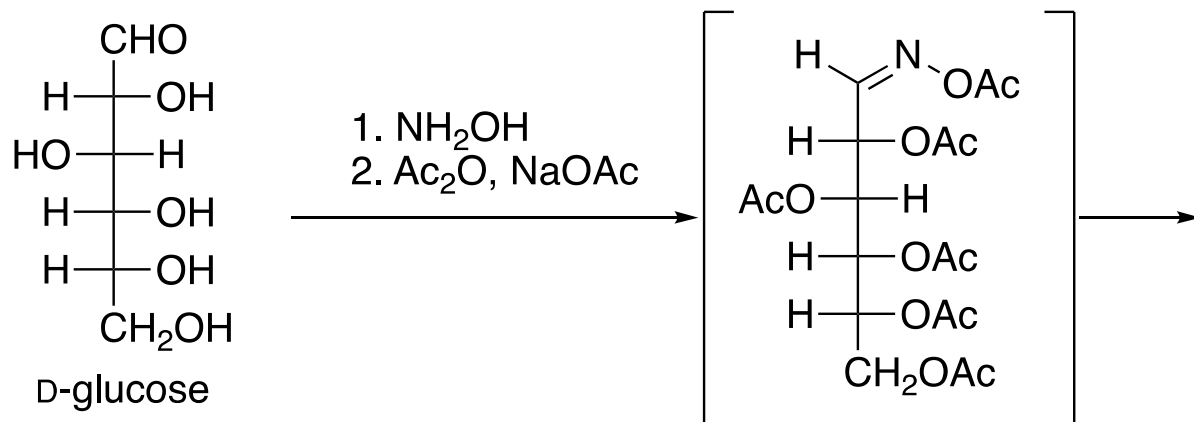
24.6 Lengthening Monosaccharides: Fischer–Kiliani Synthesis

- **Fischer-Kiliani synthesis:** aldoses can be lengthened by one carbon using nucleophilic addition of cyanide (cyanohydrin formation) followed by reduction and hydrolysis



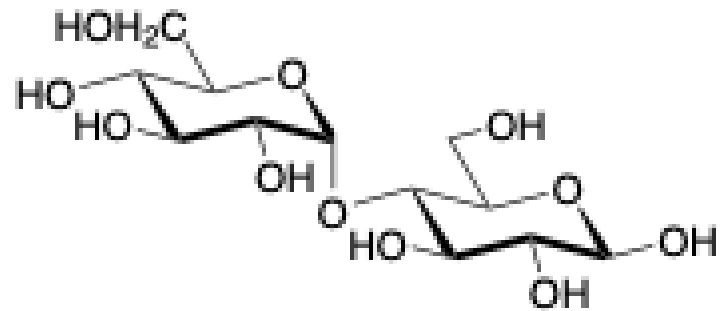
24.6 Shortening Monosaccharides: Wohl Degradation

- Wohl degradation:** aldoses can be shortened by one carbon (with loss of C-1) via conversion to a cyanohydrin followed by elimination of HCN promoted by a base

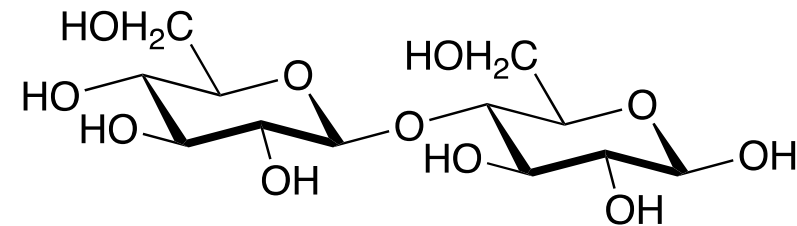


24.7 Disaccharides Contain Two Sugar Units | 1 of 2

- **Disaccharides** include two sugar units joined by a **glycosidic bond** between the anomeric carbon of one sugar and a hydroxyl oxygen in the other sugar (the latter can vary)
- Maltose is obtained from the hydrolysis of starch
 - Reducing sugar
 - Undergoes mutarotation
- Cellobiose is a related disaccharide comprising β -glucose units



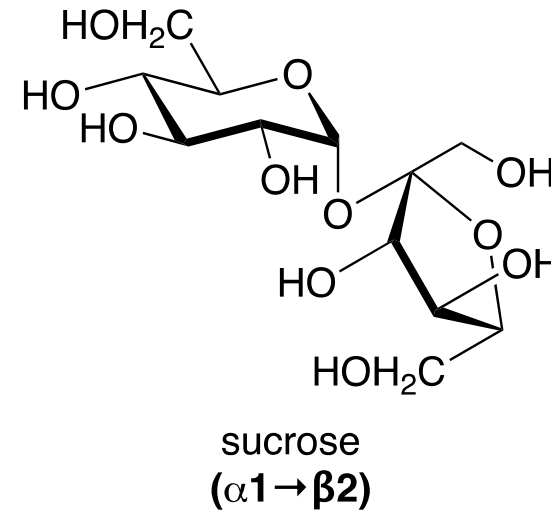
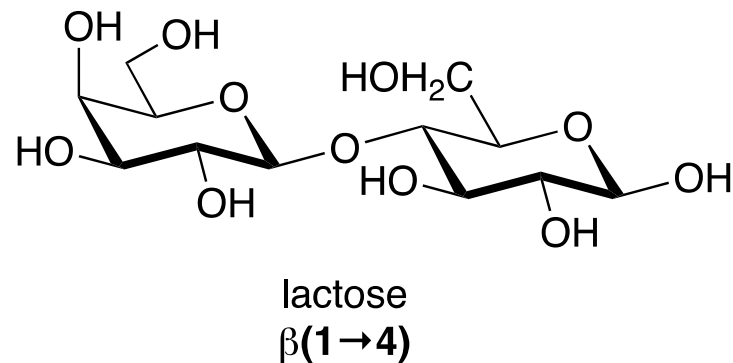
maltose
 $\alpha(1 \rightarrow 4)$



cellobiose
 $\beta(1 \rightarrow 4)$

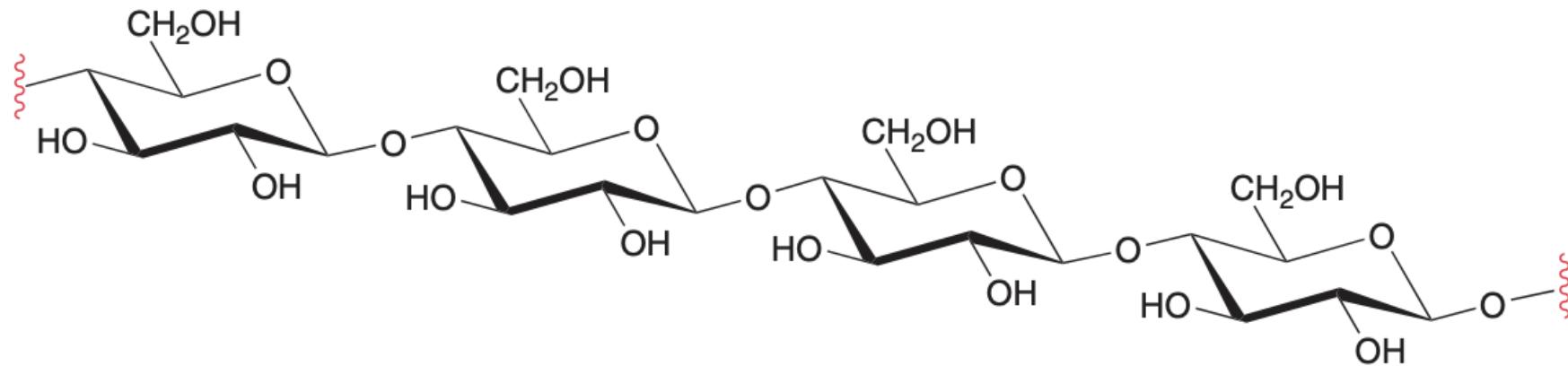
24.7 Disaccharides Contain Two Sugar Units | 2 of 2

- Lactose is a $\beta(1\rightarrow4)$ disaccharide containing two different monosaccharide units (galactose and glucose)
- Sucrose contains α -glucose in its pyranose form and β -fructose in its furanose form
 - No mutarotation and non-reducing



24.8 Polysaccharides Contain Multiple Sugar Units | 1 of 2

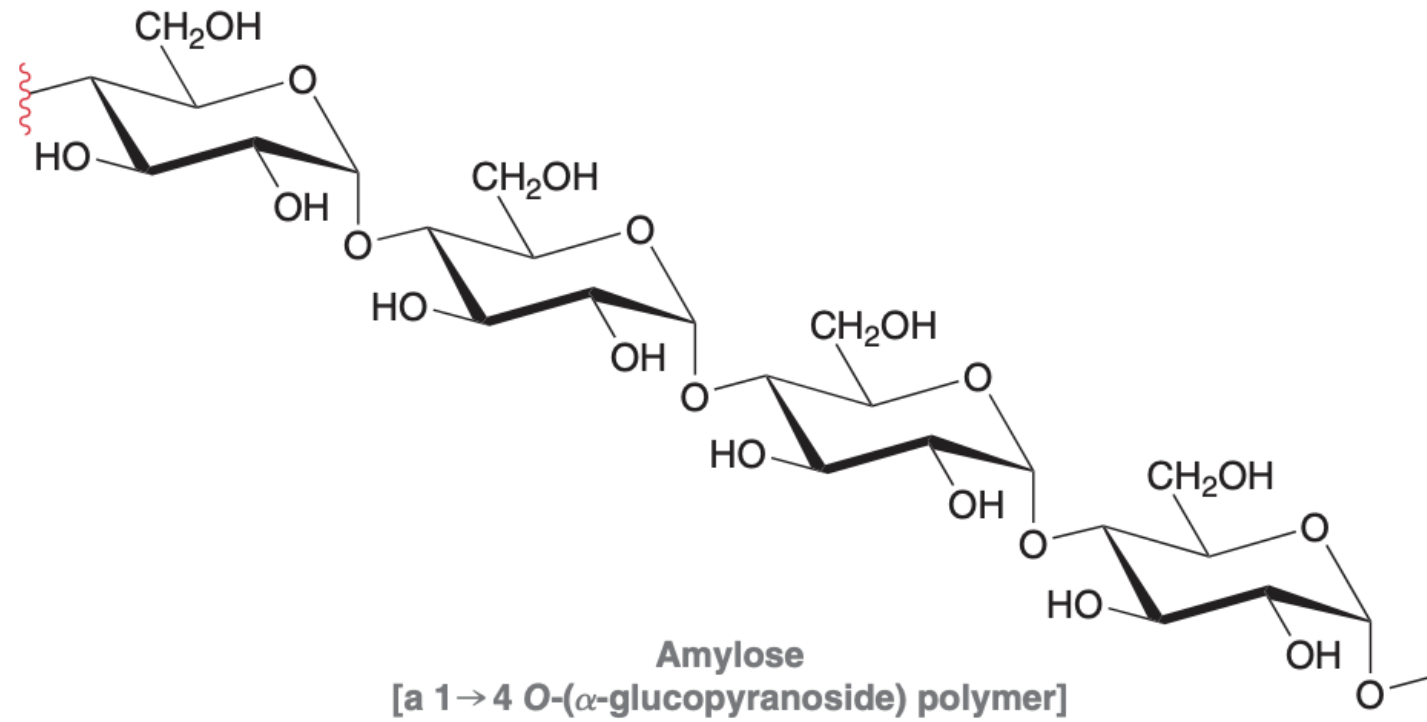
- **Polysaccharides** contain many sugar units linked by glycosidic bonds
- Formed via repeated glycosidation of C-1 on the growing polysaccharide chain
- Applied by living systems for structural rigidity, energy storage, and cell signaling
- Cellulose, a polymer of β -glucose, provides structural rigidity in plants



Cellulose
[α 1 $\rightarrow$ 4 O-(β -glucopyranoside) polymer]

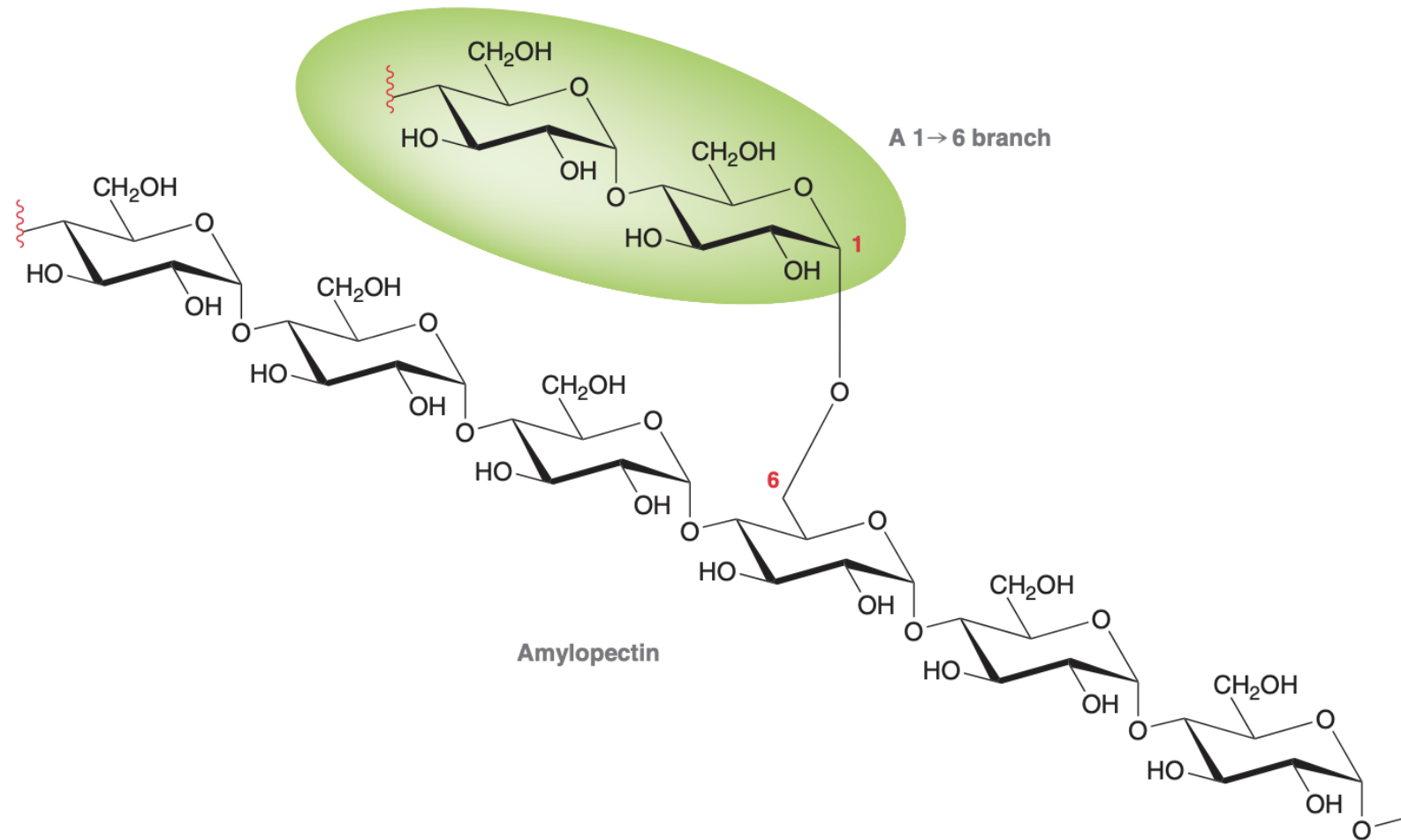
24.8 Polysaccharides Contain Multiple Sugar Units | 2 of 2

- Starches are used primarily for energy storage and are a major component of plant-based foods
- Amylose is a linear polymer of α -glucose



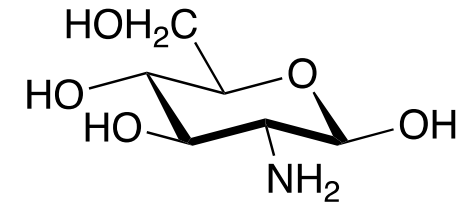
24.8 Polysaccharides Contain Multiple Sugar Units

- Starches are used primarily for energy storage and are a major component of plant-based foods
- Amylopectin includes $\alpha(1\rightarrow6)$ branches

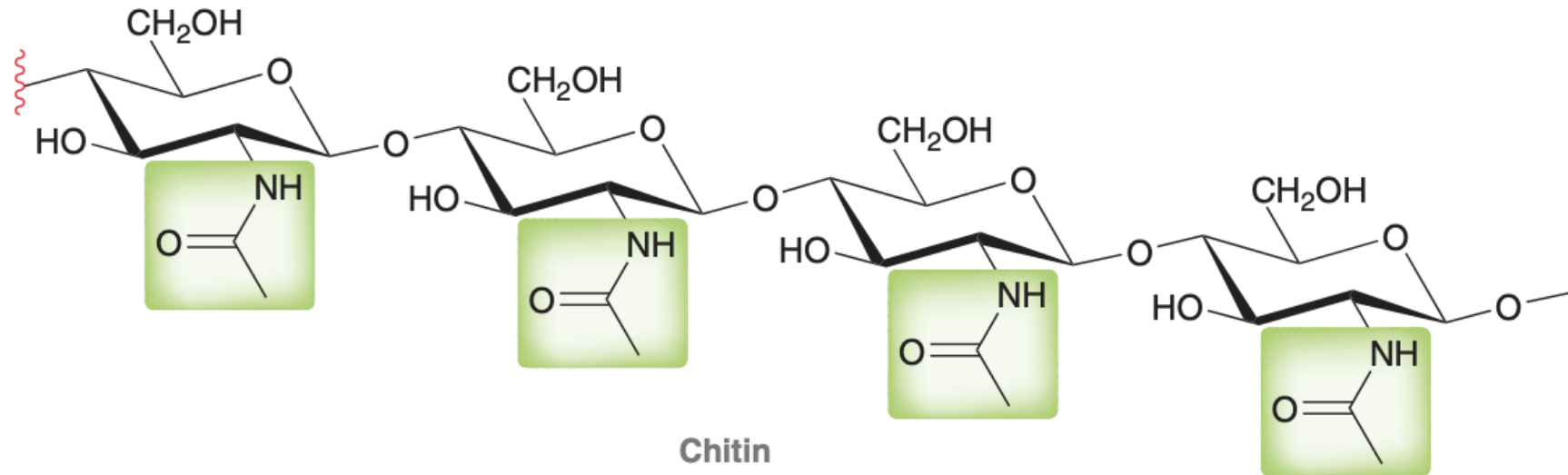


24.9 Amino Sugars

- In an **amino sugar**, one (or more) of the $-OH$ groups at a non-anomeric carbon is replaced with an amino group ($-NH_2$)
- For example, glucosamine contains an amino group linked to C-2
- Chitin is a highly rigid polymer of *N*-acetylglucosamine ([GlcNAc](#)) found in insect exoskeletons

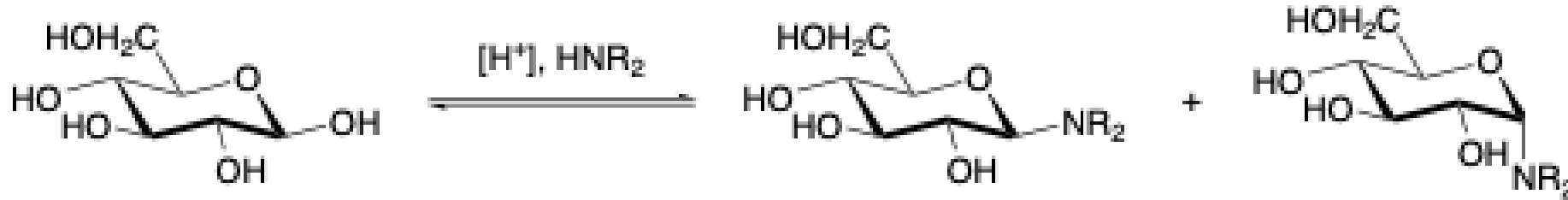


β -D-glucosamine



24.10 Nucleosides are *N*-Glycosides

- Glycosidation of an amine results in a linkage between nitrogen and C-1 and an ***N*-glycoside**



- Nucleosides** are *N*-glycosides of the nitrogenous bases with ribose (RNA) or deoxyribose (DNA)

